AI-Enabled Program Review Automation

A Human-Centered System for Translating Institutional Data into Decision-Ready Narratives

Executive Summary

Institutional research teams in higher education face a persistent challenge: translating fragmented, multi-source data into clear, consistent, and decision-ready program review reports. This process is typically manual and time-intensive.

This paper presents a practical implementation of an AI-enabled system designed to automate the generation of program review narratives from structured data sources. The system evolved from manual prompt-based workflows to a structured, reusable framework integrating prompt engineering, validation logic, and a portable execution model.

The current implementation produces complete, formatted reports with embedded quality assurance and human validation. This approach reduced report production time by approximately 50% while improving consistency and enabling institutional research staff to shift from manual reporting to higher-value analysis and decision support.

Problem Context

Program review is not a reporting problem. It is a data-to-decision translation problem.

Institutional research teams must synthesize:

- Enrollment trends
- Student demographics
- Course performance
- Completion outcomes

These data sources are often distributed across multiple systems and formats, requiring significant manual effort to interpret and communicate effectively.

Key constraints include:

- Limited staffing capacity
- Increasing demand for data-informed decision-making
- Varying data literacy among program faculty and staff
- Requirements for accuracy, consistency, and compliance

The result is a bottleneck where time is spent constructing reports rather than generating insights.

System Evolution

The solution was developed through iterative phases, each addressing specific limitations and increasing system reliability.

Phase Progression

Phase 1: Manual AI Assistance

- AI used for isolated narrative generation
- High variability and manual rework

Phase 2: Structured Prompting

- Section-based prompts and formatting rules
- Improved consistency but still required manual assembly

Phase 3: Custom GPT System

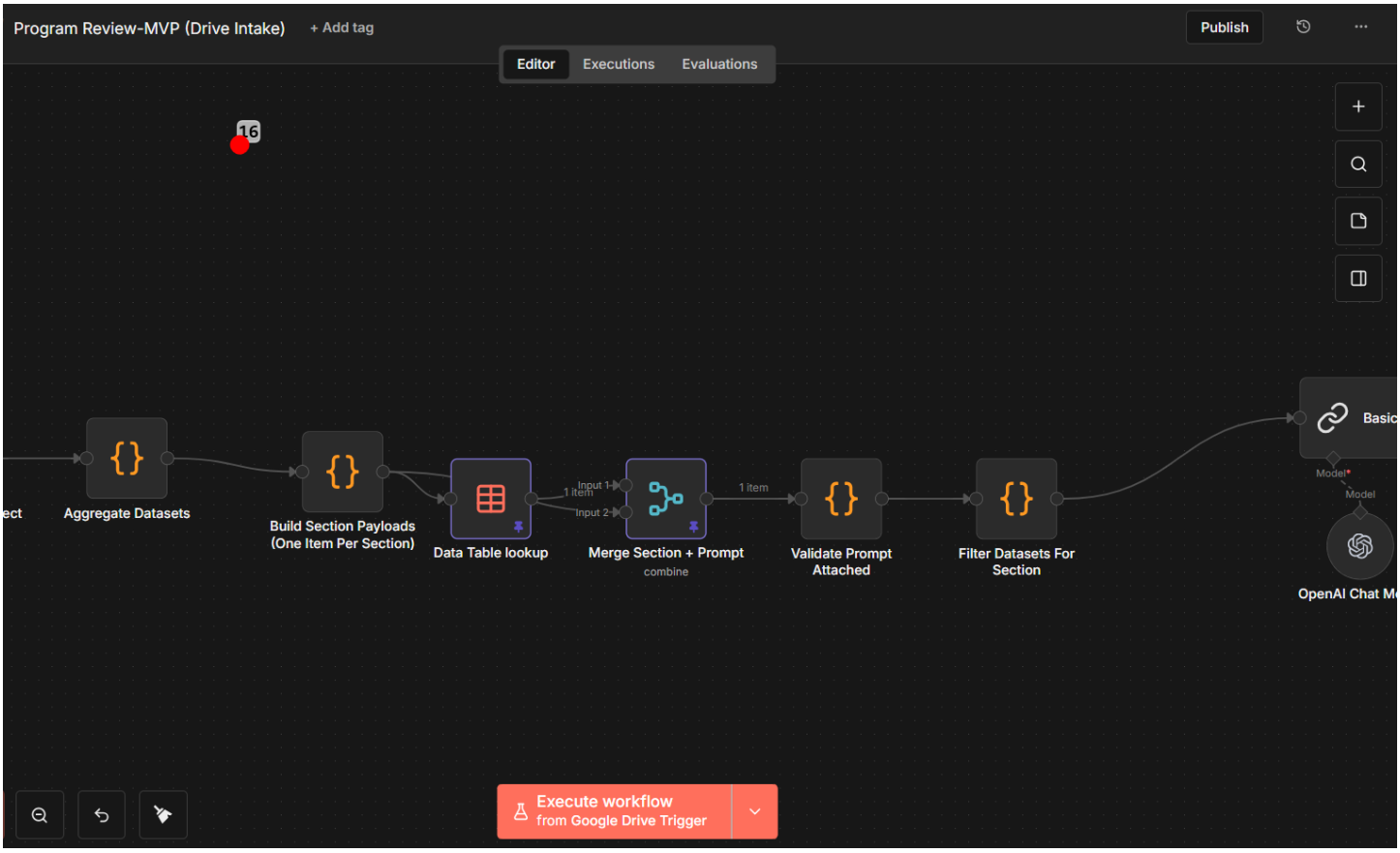
- Embedded examples and domain rules
- Introduced AI-driven analysis and QA loops

Phase 4: Persistent System Design

- Centralized prompts and logic
- Reduced iterative correction

Phase 5: Workflow Automation (n8n)

- Designed an end-to-end orchestration concept using structured data pipelines
- Identified increased scalability potential alongside significant implementation complexity
- Concluded that a lighter-weight, prompt-driven architecture would be more efficient and maintainable



Phase 6: Skill-Based Execution (Current State)

- Consolidated system into a portable prompt-driven framework (skill.md)
- Reduced infrastructure overhead while automating end-to-end analysis and report generation

Key Insight

The primary shift was from using AI tools to designing a system that integrates data processing, structured prompts, validation logic, and human oversight.

Methodology and Validation Framework

The system operates as a structured pipeline designed to ensure reliability, consistency, and compliance.

Core Methodology

Data Standardization

- Structured Excel inputs aligned to report sections
- Consistent naming and organization

Data Mapping

- Explicit or inferred assignment of datasets to report sections
- Prevents misuse of data

Pre-Processing

- Identification of table structures and metrics
- Handling of small sample sizes and totals

Prompt Architecture

Modular, section-based prompts

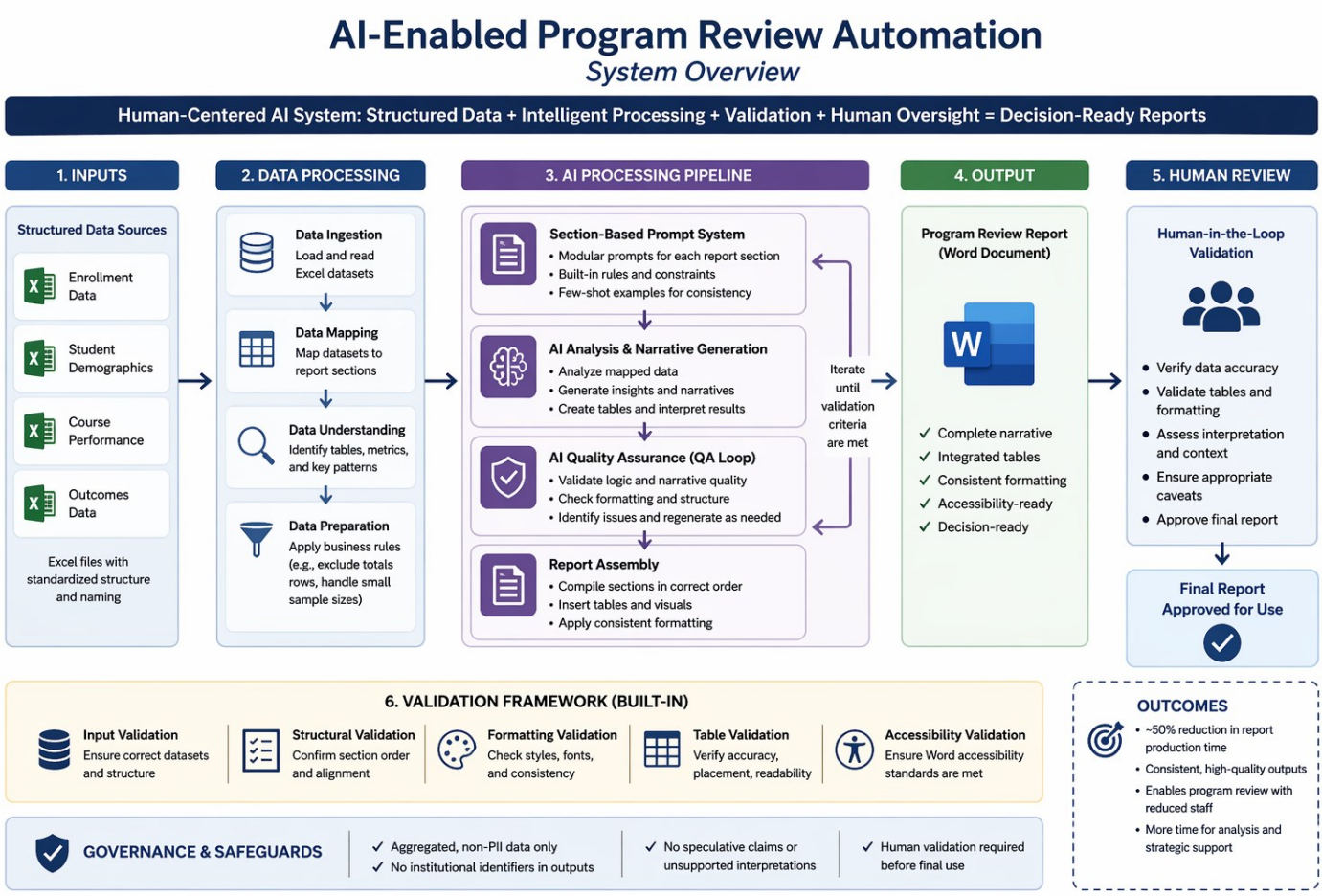
- Embedded constraints to prevent overinterpretation
- Few-shot examples to guide output quality
- Consolidated rules to improve efficiency

Validation Framework

- Input Validation – correct data sources used
- Structural Validation – proper section order and alignment
- Formatting Validation – Word-native structure and consistency
- Table Validation – correct placement and readability
- Accessibility Validation – compliance with accessibility standards
- QA Loop – iterative correction until requirements are met

Human-in-the-Loop Design

- Data accuracy verification
- Table and formatting validation
- Interpretation of trends
- Application of appropriate caveats



Results and Impact

Operational Efficiency

- Reduced report production time by approximately 50%
- Shifted effort from manual construction to validation and analysis

Scalability and Consistency

- Applied across 10+ program reports
- Standardized structure, tone, and logic

Quality and Reliability

- High-confidence outputs requiring minimal revision
- Improved through validation rules and structured prompts

Stakeholder Impact

- Program Leadership gained integrated insights
- Institutional Research enabled the introduction of program review processes despite reduced staffing capacity
- Executive Leadership responded positively

Strategic Impact

- Enabled shift from reporting to decision support

- Increased time for analysis and insight generation

Data Governance and Risk Mitigation

- Only aggregated, non-PII data used
- No institutional identifiers included
- Human validation required for all outputs
- No speculative or unsupported claims

Key Lessons Learned

- Standardized inputs are required for automation
- Prompt design is more important than model selection
- Simpler systems scale more effectively
- Few-shot examples improve output quality
- Validation must be explicit
- Human oversight is required

Conclusion

This work demonstrates that effective AI implementation is not defined by the tools used, but by the design of the system in which they operate.

By integrating structured methodology, validation logic, and human oversight, AI can function as a reliable component of institutional workflows rather than an experimental capability.

This approach provides a scalable framework for applying AI to other data-intensive processes, improving efficiency, consistency, and decision-making.